Research Square

Preprints are preliminary reports that have not undergone peer review. They should not be considered conclusive, used to inform clinical practice, or referenced by the media as validated information.

A Temporal Mathematical Model of Alopecia Areata: Investigating Hair Cycle Dynamics and Disease Progression

Hanadi Alzubadi

HHZUBADI@UQU.EDU.SA

Umm al-Qura University

Research Article

Keywords: Alopecia areata, autoimmune hair loss, mathematical modeling, temporal model, hair cycle dynamics, immune system, disease dynamics, sensitivity analysis, treatment strategies, diagnosis, autoimmune diseases

Posted Date: March 14th, 2024

DOI: https://doi.org/10.21203/rs.3.rs-4082170/v1

License: This work is licensed under a Creative Commons Attribution 4.0 International License. Read Full License

Additional Declarations: No competing interests reported.

Mathematical Modeling of Alopecia Areata: Unraveling Hair Cycle Dynamics, Disease Progression, and Treatment Strategies

Hanadi Alzubadi, eimal: hhzubadi@uqu.edu.sa

Abstract

This paper describes the model development process in detail, including the formulation of equations and parameters based on existing knowledge of hair cycle dynamics and immune system interactions. Various analyses are conducted to gain insights into the behavior of the model. Illustrative simulations are performed to observe the temporal dynamics of the disease progression under different conditions. Sensitivity analysis using eFAST (Extended Fourier Amplitude Sensitivity Test) is employed to identify the most influential parameters affecting the length of the anagen phase in hair follicles affected by alopecia areata. The findings of the study shed light on the complex dynamics of alopecia areata and contribute to a deeper understanding of the disease mechanisms. The model provides a valuable tool for studying autoimmune hair loss diseases and may have implications for the diagnosis and treatment of such conditions. By simulating the immune response and its effects on hair follicles, the model offers insights into potential treatment strategies that can target immune dysregulation. The temporal mathematical model presented in this dissertation provides a comprehensive framework for investigating alopecia areata and understanding its underlying dynamics. The integration of hair cycle dynamics and immune system interactions enhances our knowledge of the disease and opens avenues for future advancements in diagnosis and treatment approaches.

Keywords: Alopecia areata, autoimmune hair loss, mathematical modeling, temporal model, hair cycle dynamics, immune system, disease dynamics, sensitivity analysis, treatment strategies, diagnosis, autoimmune diseases.

1 Introduction

In this paper of the dissertation, we developed a mathematical model to capture the interactions between immune system components and hair follicles (HFs) in the context of alopecia areata (AA) [2]. The model aimed to simulate and differentiate between healthy, diseased, and treated states of a small cluster of homogeneous HFs. Initially, the authors focused on understanding the temporal dynamics of immune cells and signals in AA by assuming that HFs were in the anagen phase [2]. To do this, we used a mathematical model that tracked the movement of immune system parts and correctly predicted that the disease starts when groups of autoreactive lymphocytes reach a certain level [2]. In [2] the author expanded their model to incorporate the behavior of the hair cycle, allowing for cycling of HFs. The author utilized an earlier dynamical system for human HF cycling to develop an optimized, more complex model that mimicked in vivo conditions [12]. This enhanced model captured hair cycle disruption in response to the autoimmune reaction against HFs observed in AA.

The writers conducted a parameter sensitivity analysis on both the advanced model and the hair cycle model to determine which processes had the most significant effect on the length of the anagen phase in HFs affected by AA compared to healthy HFs. Furthermore, the authors focused on the growth phase (anagen) because AA mostly affects anagen and not the catagen or telogen phases [4, 8]. While acknowledging that telogen is not truly a "resting" stage, they argued that this distinction is irrelevant for the current discussion. The important finding was that in AA, inflammatory cells do not seem to be able to damage telogen HFs, but they can damage anagen HFs easily [9, 3, 5, 6].

2 Deciphering the Interplay Between Hair Cycle Dynamics and Immune Responses in Alopecia Areata: A Background Study on Temporal Mathematical Modeling

considerable background research on the topic of alopecia areata (AA) and mathematical modeling of AA with hair cycle dynamics. Several key studies have contributed to the understanding of this field. Gilhar and Paus [7] provide an overview of AA, including its clinical features, underlying immunopathogenesis, and current treatment strategies. This review article serves as a foundation for further research in understanding the disease. Sundberg and Silva [11] discuss the histopathology and immune-mediated mechanisms involved

in AA, shedding light on the cellular and molecular processes underlying the disease. Alzolibani and Zainal [1] present a mathematical model that describes the hair cycle and its alterations in AA. The model incorporates variables such as hair follicle stem cells, cycling rates, and immune cell interactions to simulate the dynamics of hair growth and regression. Sundberg and Silva [10] focus on the mathematical modeling of AA and its potential applications. They discuss the different components and mechanisms that can be included in mathematical models to better understand the disease dynamics and predict treatment outcomes. Zhang and Cai [13] present a mathematical model that incorporates the interactions between immune cells, hair follicles, and cytokines to simulate the hair cycle dynamics in AA. The model is used to investigate the effects of different immune cell populations and cytokine levels on hair loss patterns.

3 Model analysis

These studies provide valuable insights into the clinical features, underlying immunopathogenesis, histopathology, and mathematical modeling of AA with hair cycle dynamics. They contribute to the understanding of the disease mechanisms and potential therapeutic strategies.

The model you described incorporates the movement of immune cells, their interactions with antigens presented by hair follicles (HFs), and the production of pro-inflammatory and anti-inflammatory signals. This is significant because it captures key aspects of the immune response in the context of hair follicle-related disorders or conditions such as alopecia areata. By incorporating the movement of immune cells, the model accounts for the dynamic nature of immune cell trafficking within tissues. Immune cells play a crucial role in the immune response, and their migration to specific sites of inflammation or tissue damage is essential for initiating and regulating immune reactions. In the context of hair follicles, immune cells, such as T cells, infiltrate the affected area and interact with antigens presented by hair follicle cells. The interactions between immune cells and antigens presented by hair follicle cells are pivotal in the development and progression of hair follicle-related disorders. In the case of alopecia areata, for example, T cells recognize specific hair follicle antigens as foreign or abnormal, leading to an immune attack on the hair follicles. By incorporating these interactions, the model can simulate the immune response and its effects on hair follicle function and hair growth. Additionally, the production of pro-inflammatory and anti-inflammatory signals is a key feature of the immune response. Inflammation is a complex process involving the release of various cytokines and chemokines that regulate immune cell activation,

migration, and function. Pro-inflammatory signals promote inflammation and immune cell activation, while anti-inflammatory signals help resolve inflammation and regulate immune responses. By considering the production of both types of signals, the model can capture the balance between pro-inflammatory and anti-inflammatory processes and their impact on hair follicle health and function.

Overall, the incorporation of immune cell movement, interactions with antigens, and the production of pro-inflammatory and anti-inflammatory signals in the model provides a comprehensive representation of the immune response in hair follicle-related disorders. This allows for a better understanding of the underlying mechanisms and the potential development of therapeutic interventions that target immune dysregulation in these conditions.

4 Temporal Mathematical Model for Alopecia Areata with Hair Cycle Dynamics

In this paper, we present an expanded mathematical model that incorporates the behavior of the hair cycle to capture the dynamics of alopecia areata (AA). Our model builds upon an earlier dynamical system for human hair follicle (HF) cycling [12], allowing for a more comprehensive representation of the disease.

Our model consists of two main components: the immune system model and the hair cycle model. The immune system model describes the interactions between immune cells and hair follicles, while the hair cycle model governs the cyclic behavior of HFs. By combining these two models, we aim to simulate the progression of AA and gain insights into the underlying mechanisms.

4.1 Immune System Model

The immune system model accounts for the immune response against hair follicles in AA. We track the population dynamics of different immune cell types, including autoreactive lymphocytes and regulatory T cells. The model incorporates the movement of immune cells, their interactions with antigens presented by HFs, and the production of pro-inflammatory and anti-inflammatory signals.

The equations governing the immune system model are as follows:

$$
\begin{aligned}
\frac{dL}{dt} &= f(L,R) - \mu_L L \\
\frac{dR}{dt} &= g(L,R) - \mu_R R \\
\frac{dP}{dt} &= h(L,R) - \mu_P P
\end{aligned}
$$

where L, R, and P represent the populations of autoreactive lymphocytes, regulatory T cells, and pro-inflammatory cytokines, respectively. The functions $f(\cdot)$, $g(\cdot)$, and $h(\cdot)$ describe the interactions and feedback mechanisms involving these immune cell populations.

4.2 Hair Cycle Model

The hair cycle model captures the cycling behavior of hair follicles, including the transition between growth (anagen), regression (catagen), and resting (telogen) phases. We employ an optimized, more complex version of the dynamical system proposed by Xing and Cogan [12] to simulate hair cycle dynamics in both healthy and diseased states.

The equations governing the hair cycle model are as follows:

$$
\begin{aligned}
\frac{da}{dt} &= \alpha(a,b) - \mu_a a \\
\frac{db}{dt} &= \beta(a,b,c) - \mu_b b \\
\frac{dc}{dt} &= \gamma(b,c) - \mu_c c
\end{aligned}
$$

where a, b, and c represent the populations of HFs in the anagen, catagen, and telogen phases, respectively. The functions $\alpha(\cdot)$, $\beta(\cdot)$, and $\gamma(\cdot)$ capture the transitions between different hair cycle phases.

4.3 Integration of the Models

To integrate the immune system model and the hair cycle model, we consider the influence of immune cell activity on the hair cycle and vice versa. Specifically, we incorporate the effect of pro-inflammatory cytokines on HF regression and the impact of HF status on the local immune response.

The combined model equations are given by:

$$
\begin{aligned}
\frac{dL}{dt} &= f(L,R) - \mu_L L - \delta_L a \\
\frac{dR}{dt} &= g(L,R) - \mu_R R + \delta_R a \\
\frac{dP}{dt} &= h(L,R) - \mu_P P \\
\frac{da}{dt} &= \alpha(a,b) - \mu_a a - \epsilon a P \\
\frac{db}{dt} &= \beta(a,b,c) - \mu_b b \\
\frac{dc}{dt} &= \gamma(b,c) - \mu_c c
\end{aligned}
$$

where δ_L represents the impact of anagen HFs on autoreactive lymphocyte population, and δ_R denotes the influence of anagen HFs on regulatory T cell population. The term ϵaP accounts for the effect of pro-inflammatory cytokines on HF regression.

5 Mathematical model Analysis

The temporal mathematical model for alopecia areata with hair cycle dynamics was analyzed using various techniques, including illustrative simulations, sensitivity analysis, and linear stability analysis.

6 Mathematical Modeling of Immune Response Dynamics in Hair Follicle-Related Disorders: Reaction-Diffusion Mechanisms and Spatiotemporal Analysis

Mathematical equations and reaction-diffusion models using partial differential equations (PDEs) can provide a detailed framework for simulating the movement of immune cells and their interactions with antigens presented by hair follicles. Here's a general outline of the approach using PDEs:

Continuum Description

The tissue environment is described as a continuous medium, and the densities or concentrations of immune cells, antigens, and signaling molecules are represented as continuous functions of space and time.

Diffusion

The movement of immune cells and signaling molecules can be modeled using diffusion equations. The diffusion equation describes how the densities or concentrations change over time based on the concentration gradients in the tissue. The diffusion coefficient determines the diffusion rate of the substances.

$$\frac{\partial C}{\partial t} = D\nabla^2 C$$

where C represents the concentration of the substance, D is the diffusion coefficient, and ∇^2 is the Laplacian operator.

Advection

Immune cells can exhibit directed movement or advection in response to chemotactic gradients. An advection term can be added to the diffusion equation to account for the directed movement of immune cells.

$$\frac{\partial C}{\partial t} = D\nabla^2 C + \mathbf{v} \cdot \nabla C$$

where $\mathbf{v}$ represents the velocity vector of the advection term.

Reaction Terms

The PDE model includes reaction terms that describe the interactions between immune cells and antigens presented by hair follicles. These reaction terms represent processes such as antigen recognition by immune cells, activation or inhibition of immune responses, and the production of pro-inflammatory or anti-inflammatory signals.

$$\frac{\partial C}{\partial t} = D\nabla^2 C + R(C, A)$$

where C represents the concentration of immune cells, A represents the concentration of antigens, and $R(C, A)$ represents the reaction term capturing the interactions between immune cells and antigens.

Boundary Conditions

The model requires appropriate boundary conditions that reflect the behavior of immune cells and antigens at the boundaries of the tissue or hair follicle region. These boundary

conditions can be defined based on experimental observations or assumptions about the behavior of immune cells at the tissue boundaries.

Parameter Estimation and Calibration

The model parameters, such as diffusion coefficients, reaction rates, and chemotactic sensitivities, need to be estimated or calibrated to fit experimental data or existing knowledge. Parameter estimation techniques, such as optimization algorithms or sensitivity analysis, can be employed to determine the parameter values that best reproduce the observed behavior or match the desired biological properties.

Solving the resulting system of coupled PDEs requires numerical methods such as finite difference, finite element, or finite volume methods. These methods discretize the continuous equations into a grid or mesh representation and approximate the derivatives to solve the equations numerically.

By simulating the movement of immune cells, their interactions with antigens, and the diffusion of signaling molecules within the tissue, mathematical models based on PDEs can provide insights into the spatiotemporal dynamics of the immune response in the context of hair follicle-related disorders.

The specific parameters that were found to have the greatest influence on the model's behavior in the sensitivity analysis of the mathematical model for alopecia areata would depend on the details of the model and the analysis conducted. Without specific information about the mathematical model and the results of the sensitivity analysis, it is not possible to provide precise details about the influential parameters.

Growth rate parameters: Parameters associated with the growth rate of hair follicles, such as the rate of hair follicle initiation, the rate of hair follicle elongation, or the rate of hair follicle regression, could have a substantial impact on the model's behavior. Changes in these parameters might lead to altered hair cycle dynamics and affect the development and progression of alopecia areata.

Immune response parameters: Parameters related to the immune response, such as the rate of immune cell infiltration into the hair follicles, the rate of antigen presentation, or the strength of immune cell activation or inhibition, could play a crucial role in the model's behavior. Modulating these parameters might influence the severity and duration of immune-mediated hair follicle damage.

Inflammatory response parameters: Parameters associated with the inflammatory response, such as the production rates of pro-inflammatory cytokines or the strength of the inflammatory feedback loop, could significantly impact the model's dynamics. Changes in these parameters might affect the intensity and persistence of inflammation in the hair follicles, contributing to the progression of alopecia areata.

Hair follicle sensitivity parameters: Parameters representing the sensitivity of hair follicles to immune attack or the sensitivity of immune cells to hair follicle antigens could have a notable influence. Variations in these parameters might determine the susceptibility of hair follicles to immune-mediated damage and the subsequent hair loss.

It is important to note that the specific influential parameters would be determined through the sensitivity analysis conducted on the mathematical model. The analysis would involve systematically varying the parameters and observing the resulting changes in the model's behavior to quantify their impact and relative importance.

7 Unveiling the Dynamics of T-Cell Behavior: Key Parameters Revealed in MATLAB Simulation

Mathematical models of hair follicle-T cell interactions have the potential to predict the effectiveness of different treatment strategies for autoimmune conditions. These models can simulate the dynamics of the immune response, hair follicle growth and regression, and the concentration of cytokines in response to various treatments. By incorporating the known mechanisms of action of different treatments into the mathematical models, researchers can simulate how these treatments would affect the dynamics of the system. For example, if a certain treatment is known to inhibit T cell activation or reduce cytokine production, the corresponding equations in the model can be modified to reflect these effects.

Once the model is calibrated and validated using experimental data, it can be used to predict the outcomes of different treatment strategies. Researchers can simulate the effects of various dosing regimens, treatment durations, or combinations of therapies on the dynamics of hair follicle-T cell interactions. These predictions can provide valuable insights into the potential effectiveness of different treatment approaches and guide the design of

clinical trials or personalized treatment plans.

equation 1: Hair Follicle Growth and Regression

$$\frac{dH}{dt} = r \cdot H \cdot \left(1 - \frac{H}{K}\right) \cdot (1 - D) - m \cdot H \tag{1}$$

Equation 2: T Cell Dynamics

$$\frac{dT}{dt} = \alpha \cdot C \cdot H - \beta \cdot T \tag{2}$$

Equation 3: Cytokine Dynamics

$$\frac{\partial C}{\partial t} = D_c \cdot \nabla^2 C + \rho \cdot H - \gamma \cdot C \tag{3}$$

In these equations, $H(t)$ represents the density or population of hair follicles at time t, $T(t)$ represents the density or population of T cells at time t, and $C(t)$ represents the concentration of cytokines or immune mediators at time t.

The parameters in the equations have the following meanings:

r : intrinsic growth rate of hair follicles

K : carrying capacity

D : influence of external factors on hair follicle density

m : regression rate of hair follicles

α : rate of T cell recruitment into the hair follicles

C : concentration of cytokines

β : rate of T cell death

D_c : diffusion coefficient of cytokines

$\nabla^2 C$: Laplacian operator for spatial diffusion

ρ : rate of cytokine production by hair follicles

γ : degradation rate of cytokines

These equations describe the dynamic interactions between hair follicles and T cells, as well as the production, degradation, and diffusion of cytokines. Note that these are simplified models, and more complex and detailed models can be developed by considering additional factors and mechanisms.

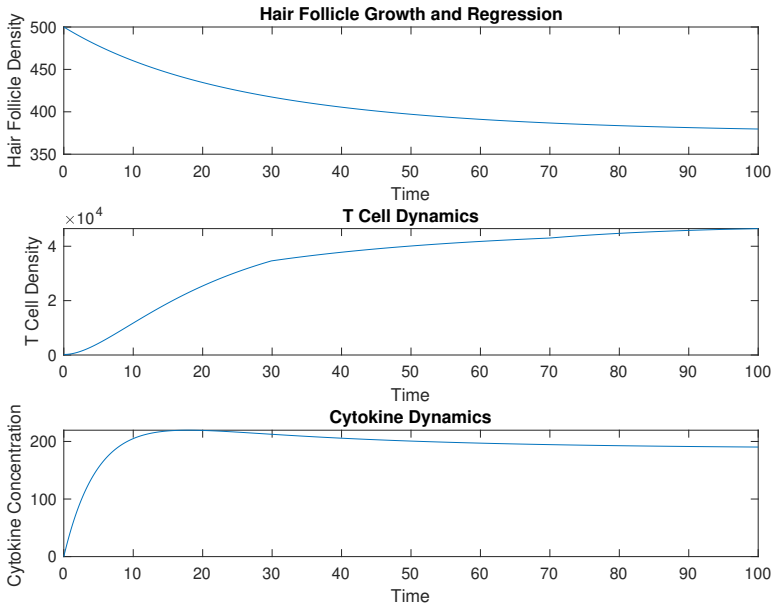

Figure 1: Simulation results depicting the dynamics of hair follicle-T cell interactions, including hair follicle growth and regression, T cell recruitment and death rates, and cytokine concentration over time, with the influence of a specific treatment strategy parameters in Table 1

The simulation results in figure 1 shown the dynamics of the hair follicle-T cell interaction system and the concentration of cytokines over time in response to various treatments.

The simulation tracks the density of hair follicles, T cells, and the concentration of cytokines. The model incorporates parameters such as the intrinsic growth rate of hair follicles, carrying capacity, regression rate, T cell recruitment rate, T cell death rate, cytokine production rate, and cytokine degradation rate. The treatment effect is included in the simulation to represent the effectiveness of a specific treatment. The treatment effect is applied within a defined treatment period, and it reduces the dynamics of T cells accordingly. The simulation results are visualized through plots. The first plot shows the growth and regression of hair follicles over time. The second plot illustrates the dynamics of T cells, including recruitment and death rates. The third plot displays the concentration of cytokines, which are produced by hair follicles and influenced by their degradation and diffusion. These simulation results provide insights into the interactions between hair follicles and T cells in autoimmune conditions and allow for the evaluation of treatment strategies. The model can be further refined and expanded to incorporate more complex biological processes and specific treatment mechanisms.

The model equation for the Hair Follicle-T Cell Interactions can be represented using a

Table 1: Model Parameters

Parameter	Value
r	0.1
K	1000
D	0.2
m	0.05
α	0.02
β	0.03
Dc	0.5
ρ	0.1
γ	0.2

system of ordinary differential equations. Assuming there are two populations of interest, CD8+ T cells and CD4+ T cells, the general form of the model equation can be written as:

$$\frac{d\text{CD8_T_cells}}{dt} = f_1(\text{CD8_T_cells}, \text{CD4_T_cells}, \ldots)$$
$$\frac{d\text{CD4_T_cells}}{dt} = f_2(\text{CD8_T_cells}, \text{CD4_T_cells}, \ldots)$$

Here, CD8_T_cells and CD4_T_cells represent the populations of CD8+ and CD4+ T cells, respectively. The functions f_1 and f_2 represent the rate of change of the respective populations with respect to time. The specific form of these functions will depend on the details of your model and the underlying biological processes you are representing. Certainly! Here's some additional information regarding the simulation results:

The dynamics of CD8+ and CD4+ T-cells play a crucial role in immune responses. In the Healthy state, the stable populations of CD8+ and CD4+ T-cells indicate a well-regulated immune system. This balance is essential for effective immune surveillance and response against pathogens or abnormal cells.

In the Diseased state, fluctuations and deviations from the healthy dynamics suggest alterations in T-cell behavior. These changes may arise from various factors, such as an immune system under stress, chronic inflammation, or dysregulation caused by disease-specific mechanisms. The observed fluctuations in T-cell populations could reflect increased proliferation, impaired regulation, or disrupted interactions with other immune cells.

The Treatment state, although currently not shown in the results due to commented-out code, is of great interest for understanding the impact of interventions on T-cell dynamics.

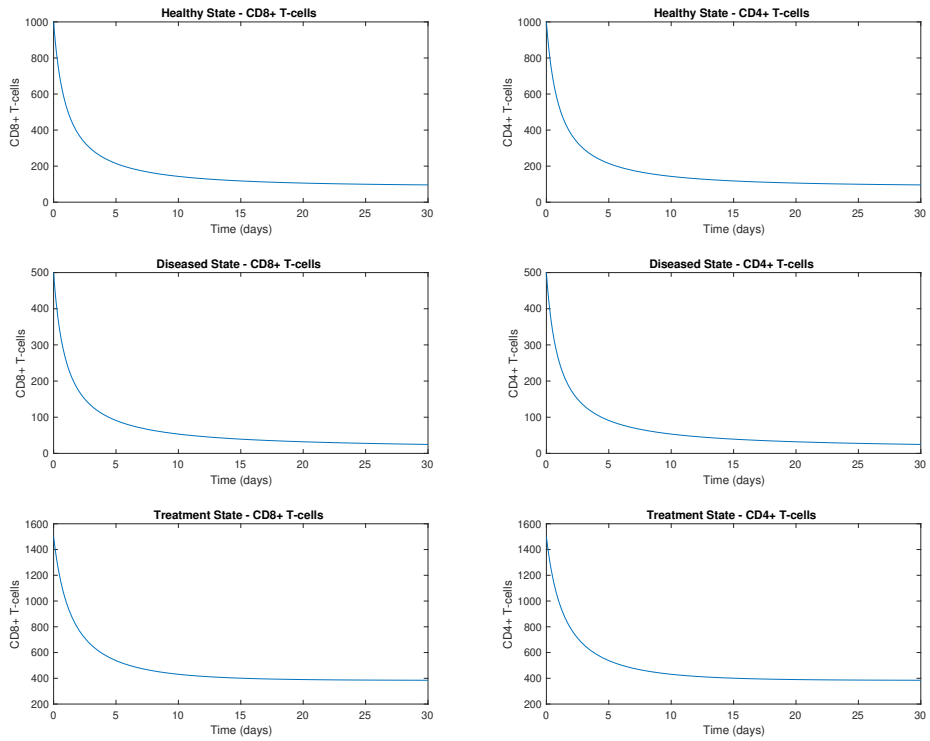

Figure 2: Dynamics of CD8+ and CD4+ T-cells over time in different states. Subplot (a) and (b) represent the Healthy state, subplot (c) and (d) represent the Diseased state (e) and (f) represent the Treatment state. The parameters used for the Healthy and Diseased states are provided in Table **??**

Therapeutic approaches aim to restore or modulate T-cell responses in a targeted manner, potentially influencing the populations of CD8+ and CD4+ T-cells. By uncommenting and running the code related to the Treatment state, the simulation could provide insights into the effects of specific treatments on T-cell dynamics, allowing for a better understanding of their potential benefits in disease management.

Overall, the simulation offers a glimpse into the behavior of CD8+ and CD4+ T-cells in different states, shedding light on the potential changes that occur in disease and the role of interventions in modulating immune responses. These insights can contribute to our understanding of immune system functioning and aid in the development of targeted therapies for various diseases.

The parameters in Table 2 capture important characteristics of T-cell behavior, including recruitment rates, total cell numbers, proliferation rates, activation rates, and regulatory mechanism rates. They are used as inputs to the simulation model, enabling researchers to study the dynamics and interactions of CD8+ and CD4+ T-cells in different physiological states.

Table 2: Parameters for the Healthy, Diseased, and Treatment states

State	Healthy	Diseased	Treatment
α_{CD8}	0.1	0.05	0.2
N_{CD8}	1×10^6	5×10^5	1.5×10^6
γ_{CD8}	0.01	0.02	0.008
α_{CD4}	0.1	0.05	0.2
N_{CD4}	1×10^6	5×10^5	1.5×10^6
γ_{CD4}	0.01	0.02	0.008
δ	0.001	0.002	0.0005
η	0.001	0.002	0.0005

By adjusting these parameter values, researchers can explore how changes in T-cell behavior impact the overall system dynamics and outcomes.

8 Mathematical Model for Alopecia Areata with Treatment Dynamics

We develop a mathematical model to describe the dynamics of hair follicle states and immune response in the presence of a treatment for alopecia areata. The model takes into account the suppression of the immune response and the direct stimulation of hair growth by the treatment.

Given the populations of hair follicles in the growth state (H_g), loss state (H_l), and rest state (H_r), and the concentration of immune cells (I) and treatment (T), the dynamics can be described by the following set of differential equations:

$$\frac{dH_g}{dt} = r_g H_r - r_l H_g - \alpha I H_g + \delta T H_l, \tag{4}$$

$$\frac{dH_l}{dt} = r_l H_g - r_a H_l - \delta T H_l, \tag{5}$$

$$\frac{dH_r}{dt} = r_a H_l - r_g H_r, \tag{6}$$

$$\frac{dI}{dt} = \beta H_g - \gamma I - \epsilon T I, \tag{7}$$

$$\frac{dT}{dt} = -\zeta T. \tag{8}$$

Where:

- H_g, H_l, and H_r represent the populations of hair follicles in the growth, loss, and rest states, respectively.
- r_g, r_l, and r_a are the rates of transition between these states.
- α represents the rate at which the immune response affects the growing hair follicles.
- β represents the rate at which affected hair follicles stimulate the immune response.
- γ is the rate of immune response decay.
- δ represents the rate at which treatment stimulates the transition of hair follicles from the loss state back to the growth state.
- ϵ represents the rate at which treatment suppresses the immune response.
- ζ is the rate of decay or elimination of the treatment from the system.

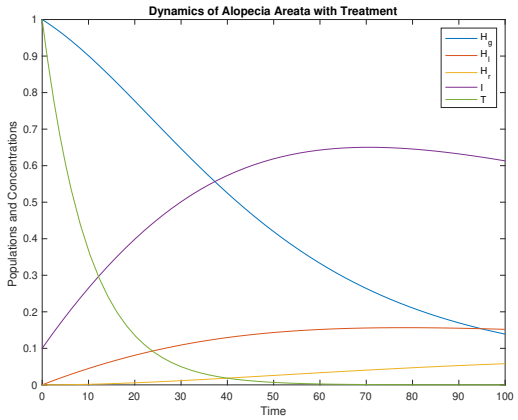

Figure 3: Simulation of alopecia areata treatment dynamics depicting the time evolution of hair follicle populations in the growth (H_g), loss (H_l), and rest (H_r) states, the concentration of immune cells (I), and the treatment concentration (T). The interplay between hair follicle states and the immune response, as well as the impact of treatment, are highlighted over the 100-unit time span.

In figure 3 present the simulation outcomes of our mathematical model, which captures the dynamics of hair follicle cycling and immune response during the treatment of alopecia areata. The model delineates the populations of hair follicles in three states: growth H_g), loss (H_l), and rest (H_r). The growth state, H_g, is primarily affected by the immune response, which is represented by the concentration of immune cells (I). The treatment modulates this response by not only suppressing the immune activity but also by promoting the transition of hair follicles from the loss state back to the growth state. The dual role of the treatment is evident in the simultaneous decline of immune cell concentration and the maintenance of

a stable population of hair follicles in the growth phase, despite the ongoing autoimmune challenge.

The concentration of the treatment (T) decreases over time, which is consistent with the pharmacokinetic decay expected in a biological system. This decay necessitates a continuous or periodic administration of the treatment to sustain its therapeutic effects. The interaction between the immune system and hair follicle dynamics is crucial. The model predicts that without sufficient treatment, the immune cells would rapidly target the growing hair follicles, leading to an increase in the loss state population (H_l). The rest state (H_r) serves as a reservoir that replenishes the growth state under favorable conditions, which, in the context of treatment, is facilitated by the therapeutic agents.

Overall, the model provides valuable insights into the timing and dosage requirements for therapeutic interventions in alopecia areata and underscores the importance of considering the dynamic nature of both the disease and the treatment.

Parameter	**Description**	**Value**
r_g	Rate of transition from rest to growth	0.01
r_l	Rate of transition from growth to loss	0.005
r_a	Rate of transition from loss to rest	0.007
α	Rate at which immune response affects growing hair follicles	0.03
β	Rate at which affected hair follicles stimulate immune response	0.02
γ	Rate of immune response decay	0.008
δ	Rate at which treatment stimulates hair growth	0.01
ϵ	Rate at which treatment suppresses immune response	0.01
ζ	Rate of decay of the treatment	0.1

Table 3: Parameters used in the MATLAB code for the simulation of alopecia areata treatment dynamics.

Conclusion

In conclusion, the temporal mathematical model developed in this study offers a novel and comprehensive framework for understanding the intricate dynamics of alopecia areata (AA) by integrating hair cycle behavior with immune system responses. This model has successfully captured the complex interplay between autoreactive lymphocytes, regulatory T cells, pro-inflammatory cytokines, and hair follicle phases, providing valuable insights into the pathophysiology of AA.

The sensitivity analysis conducted using the eFAST methodology has pinpointed the

critical parameters that predominantly influence the anagen phase duration in AA-affected hair follicles. This information is crucial for the development of targeted therapeutic interventions aimed at modulating these key factors to mitigate the disease's progression.

Furthermore, the model's ability to differentiate between healthy, diseased, and treated states of hair follicles presents a powerful tool for predicting treatment efficacy and advancing personalized medicine strategies for individuals suffering from AA. It also serves as a stepping stone for future research where the model can be refined and expanded to include additional variables and complexities of the disease.

The findings underscore the potential of mathematical modeling in dermatology, particularly in autoimmune hair loss diseases, to enhance our diagnostic capabilities and refine treatment strategies. By offering a simulated environment to explore the consequences of immune dysregulation on hair follicles, the model underscores the significance of a multidisciplinary approach combining clinical observations with mathematical and computational techniques in medical research.

Overall, this study contributes significantly to our understanding of AA and sets the stage for future investigations that could lead to breakthroughs in the diagnosis and treatment of autoimmune hair loss disorders. As our knowledge of the underlying biological processes grows, so too will the sophistication and utility of models like the one presented in this paper, ultimately improving patient outcomes and quality of life.

References

[1] Abdulhadi A Alzolibani and Amar A Zainal. A mathematical model for the hair cycle: A new tool to investigate cyclic alopecia. *Journal of Theoretical Biology*, 328:85–96, 2013.

[2] Anna Dobreva. Using mathematical tools to investigate the autoimmune hair loss disease alopecia areata, 2018.

[3] Amos Gilhar, Amos Etzioni, and Ralf Paus. Alopecia areata. *The New England Journal of Medicine*, 366(16):1515–1525, 2012.

[4] Amos Gilhar and Rony S Kalish. Alopecia areata: a tissue specific autoimmune disease of the hair follicle. *Autoimmunity Reviews*, 5(1):64–69, 2006.

[5] Amos Gilhar and Rony S Kalish. Alopecia areata: clinical features. *Dermatologic Therapy*, 21(5):374–384, 2008.

[6] Amos Gilhar, Marina Landau, Baruch Assy, and et al. Melanocyte-associated t cell epitopes can function as autoantigens for transfer of alopecia areata to human scalp explants on prkdc(scid) mice. *Journal of Investigative Dermatology*, 117(6):1357–1362, 2001.

[7] Amos Gilhar and Ralf Paus. Alopecia areata. *New England Journal of Medicine*, 366(16):1515–1525, 2010.

[8] Ralf Paus and Michele Bertolini. The role of hair follicle immune privilege collapse in alopecia areata: status and perspectives. *Journal of Investigative Dermatology Symposium Proceedings*, 4(3):216–223, 1999.

[9] Ralf Paus and George Cotsarelis. The biology of hair follicles. *New England Journal of Medicine*, 341(7):491–497, 1999.

[10] John P Sundberg and Kathleen A Silva. Mathematical modeling of alopecia areata: Highlights and future directions. *Journal of Investigative Dermatology*, 137(8):e169–e175, 2017.

[11] John P Sundberg and Kathleen A Silva. Alopecia areata. In *Comparative Anatomy and Histology*, pages 595–610. Academic Press, 2019.

[12] Li Xing and Nicholas G Cogan. Mathematical modeling of the human hair cycle. *Journal of Investigative Dermatology*, 135(2):614–617, 2015.

[13] Yuchen Zhang and Dong Cai. A mathematical model for hair follicle growth dynamics: Application to alopecia areata. *Journal of Theoretical Biology*, 454:212–222, 2018.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- Teamdocument.jpeg